

CDI LABS

Autoantibody Seromics

Empowering discoveries in disease research

BROCHURE

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What is Autoantibody Seromics?

Autoantibody seromics refers to the systematic study of autoantibodies present in the blood serum of individuals. Autoantibodies are antibodies that target the body's own tissues and cells, potentially leading to autoimmune diseases. By analyzing autoantibody profiles, researchers aim to uncover biomarkers that can aid in the early detection and monitoring of autoimmune and other diseases, as well as identify potential targets for personalized therapies aimed at modulating the immune response. Additionally, autoantibody seromics can contribute to our understanding of the underlying immunological processes involved in disease, paving the way for the development of novel treatment strategies.

Our Immune System is Adaptive and Ever-changing

Everyone's autoantibody signature is unique as an individual's immune system is informed by a lifetime of exposure to microbes, foods, vaccines, environmental factors, etc. The immune system is made up of hundreds of billions of independent immune cells competing to seek, destroy, and remember antigen targets across our lifetime.



Foods

directly provide antigens and sculpt what grows in the microbiome



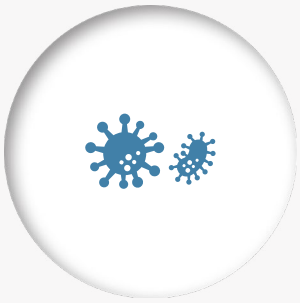
Infections and vaccines

provide TLRs and prime immunity



Activity and lifestyle

determine stress hormones and alter tolerogenic DAMPs



Our microbiome

sculpts immunity via TLRs and cross-reactive antigens



Genetics, sex and family

predispose us to certain types of immunity

A lifetime of immune experiences drives the body to make countless antibody clones.

This creates an antigen-specific antibody fingerprint that's more unique than the genome.

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Autoantibodies – a Powerful New Generation of Biomarkers

Autoantibodies are typically associated with autoimmune diseases; however, autoantibodies also impact non-autoimmune diseases, and their presence can be indicative of certain underlying processes or dysfunctions in the body. Autoantibody biomarkers in non-autoimmune diseases can offer diagnostic, prognostic, mechanistic, and therapeutic insights, making them important to understanding and managing a variety of disease conditions.

Understanding Patheogeneis

Investigating the presence of autoantibodies in non-autoimmune diseases can provide insights into the underlying mechanisms of the disease. It can help researchers and clinicians understand how the immune system interacts with various tissues and organs in the context of different diseases.

Prognostic Value

The presence of autoantibodies in non-autoimmune diseases has the potential to provide information about disease prognosis or severity. For example, the presence of certain autoantibodies in cancer patients may be associated with specific tumor characteristics, treatment outcomes or the likelihood of adverse reactions from immunotherapy.

Diagnostic Utility

Detecting autoantibodies can sometimes aid in the diagnosis of certain non-autoimmune diseases. For example, certain types of autoantibodies are associated with specific neurological disorders, such as paraneoplastic syndromes, which are disorders triggered by an immune response to cancer elsewhere in the body.

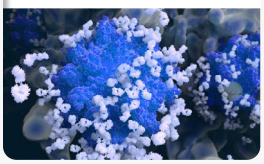
Therapeutic Targets

In some cases, autoantibodies associated with non-autoimmune diseases may serve as potential therapeutic targets. Understanding the role of these autoantibodies in disease pathogenesis could lead to the development of targeted therapies aimed at modulating the immune response or neutralizing the harmful effects of the autoantibodies.

Monitoring Disease Progression

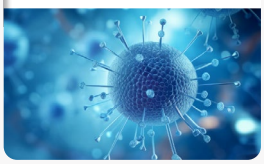
Monitoring changes in autoantibody levels over time in patients with non-autoimmune diseases can provide valuable information about disease progression or response to treatment. This can help clinicians tailor treatment strategies and monitor patients for potential complications.

Autoimmune Disease



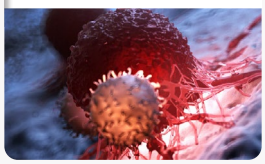
Autoantibodies play a crucial role in the development and progression of autoimmune diseases which occur when the immune system mistakenly targets and attacks the body's own tissues, mistaking them for foreign invaders.

Infectious Disease



Autoantibodies can sometimes be associated with infectious diseases, and their presence or induction is a complex aspect of the host immune response. Infections have the capability of inducing a new autoimmune response or worsening an existing condition.

Immuno-oncology



Cancer immunotherapy aims to harness the power of the immune system to target and destroy cancer cells. While autoantibodies are traditionally associated with autoimmune diseases, their role in cancer is a complex and evolving area of research.

Neurological Disease



Autoantibodies can play a significant role in neurological diseases, contributing to autoimmune disorders that affect the central nervous system. In these conditions, the immune system mistakenly targets components of the nervous system, leading to inflammation, tissue damage, and neurological symptoms.

Genetic Disease



The adaptive immune system, representing approximating half of all nucleated cells in the human body, has behavior that is particularly sensitive to genetic dysregulation. These effects can create over-active immunity with increased rates of autoimmune and inflammatory disease, or under-active immunity that makes the affected patients more susceptible to infection.

HuProt™ Human Proteome Microarray

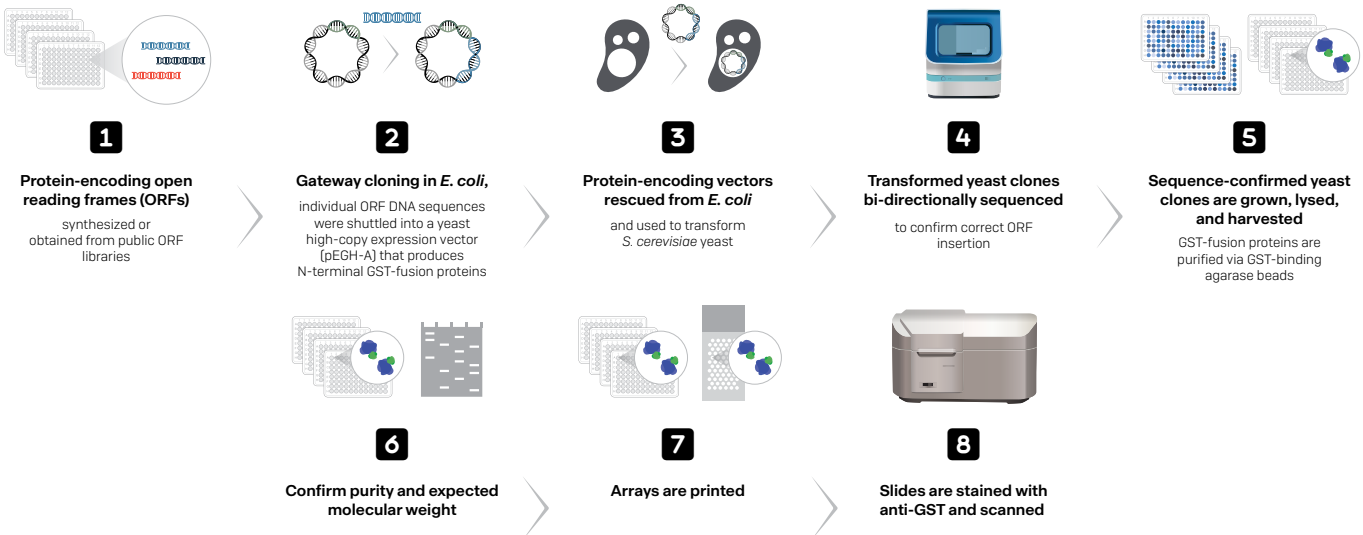
The Most Powerful Tool Ever Created for Autoantibody Profiling

HuProt consists of >21,000 unique human proteins and isoform variants – covering 16,794 unique genes. This includes 15,889 of the 19,613 canonical human proteins described in the Human Protein Atlas, with broad coverage across protein subclasses.

Most fluids contain antibodies and we can analyze them (*i.e.*, peripheral serum/plasma, cerebrospinal fluid, tissue lysates, stool, saliva), or in a synthesized sample, for any isotype (i.e., IgM, IgG, IgA, IgD, IgE). Secondary antibody binding is then quantified using fluorescence intensity measurements of features on the array. We'll provide you with those RAW readouts with normalized readouts and group-comparison statistics (optional). HuProt arrays also allow for pan-isotype autoantibody biomarker profiling versus properly folded, three-dimensional human proteins.

HuProt Library Curation and Production

There's more to HuProt than serum profiling. We've profiled nucleotide binding, antibody specificity, small molecule binding, protein-protein interactions, and more. If you can dream up a study, we can help you get it done. HuProt contains >21,000 unique proteins and was built during the NIH Protein Capture Reagents program.



1 **Protein-encoding open reading frames (ORFs)**
synthesized or obtained from public ORF libraries

2 **Gateway cloning in *E. coli*,**
individual ORF DNA sequences were shuttled into a yeast high-copy expression vector (pEGH-A) that produces N-terminal GST-fusion proteins

3 **Protein-encoding vectors rescued from *E. coli***
and used to transform *S. cerevisiae* yeast

4 **Transformed yeast clones bi-directionally sequenced**
to confirm correct ORF insertion

5 **Sequence-confirmed yeast clones are grown, lysed, and harvested**
GST-fusion proteins are purified via GST-binding agarase beads

6 **Confirm purity and expected molecular weight**

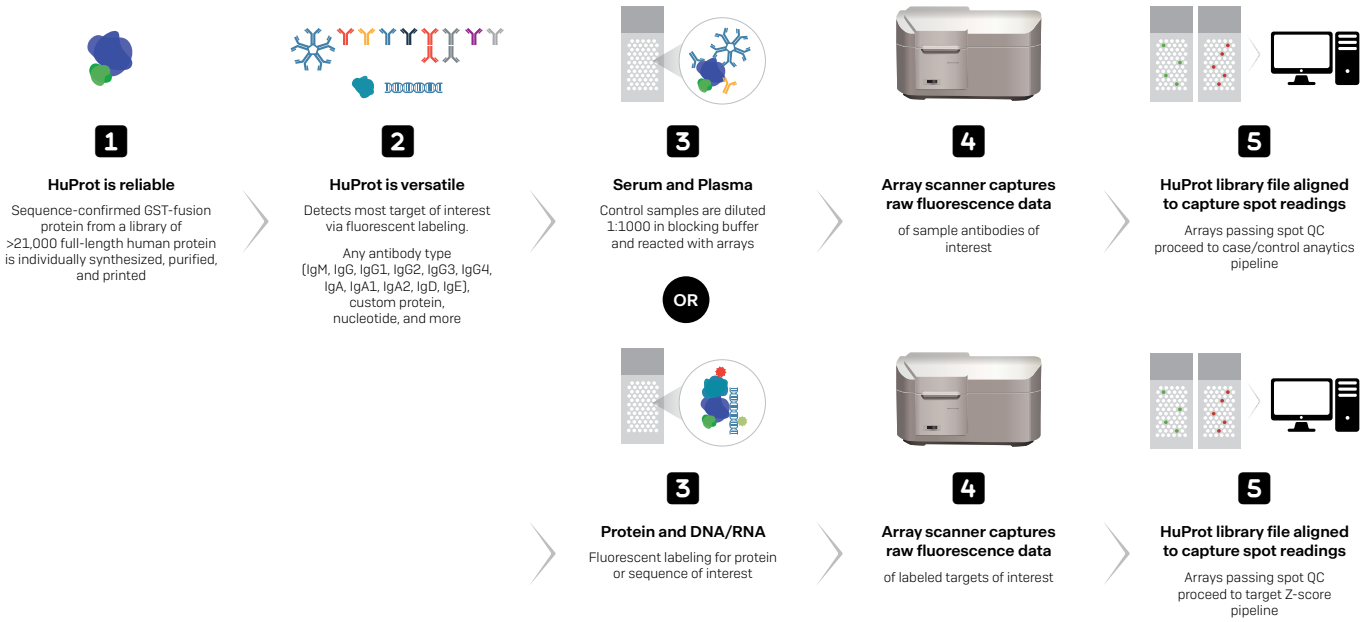
7 **Arrays are printed**

8 **Slides are stained with anti-GST and scanned**

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HuProt Library Curation and Production

HuProt is an incredibly sensitive antigen matrix platform that provides quantitative binding interaction data from a few microliters of sample. It has been used to identify everything from the immunologic fingerprint of cancer patients destined for checkpoint blockade toxicity to novel protein-protein interactions.



1

HuProt is reliable

Sequence-confirmed GST-fusion protein from a library of >21,000 full-length human protein is individually synthesized, purified, and printed

2

HuProt is versatile

Detects most target of interest via fluorescent labeling.

Any antibody type (IgM, IgG, IgG1, IgG2, IgG3, IgG4, IgA, IgA1, IgA2, IgD, IgE), custom protein, nucleotide, and more

3

Serum and Plasma

Control samples are diluted 1:1000 in blocking buffer and reacted with arrays

4

Array scanner captures raw fluorescence data

of sample antibodies of interest

5

HuProt library file aligned to capture spot readings

Arrays passing spot QC proceed to case/control anaytics pipeline

OR

3

Protein and DNA/RNA

Fluorescent labeling for protein or sequence of interest

4

Array scanner captures raw fluorescence data

of labeled targets of interest

5

HuProt library file aligned to capture spot readings

Arrays passing spot QC proceed to target Z-score pipeline

Technical Details

With over 21,000 human proteins and protein isoforms, CDI Labs' HuProt Human Proteome Microarray is the most comprehensive human full-length protein array ever created. This covers over 81% of human proteins in each major functional category of the proteome, as defined by the Human Protein Atlas, and allows hundreds of interactions to be profiled at once in a wide range of applications, including mapping antigen-specific immunity (multi-isotype serum profiling), determining monoclonal antibody specificity, and to study protein-protein interaction, substrate identification, protein-DNA binding, protein-RNA binding, and binding of some small molecules.

The HuProt expression library was created by inserting full-length human open reading frames (ORFs) inserted into a yeast high-copy expression vector that produces GST-His6 fusion proteins when induced in yeast. Each human ORF expression vector was verified for integrity by Sanger sequencing. These sequence-verified expression vectors encoding human ORFs were transformed into yeast to create our master library; this eukaryotic expression system ensures HuProt™ proteins maintain both function and proper conformation.

Full-length synthesized proteins are purified via their GST-His6 fusion tags. These proteins are then frozen in aliquots in a 384-well format for later use in printing on HuProt microarrays.

To print individual batches of HuProt arrays, the purified human proteins are then printed as duplicate spot pairs using an Arrayjet UltraMarathon printer (Arrayjet, UK) on PATH nitrocellulose slides (GraceBio, USA). Controls on HuProt include titrated GST protein, titrated human IgG, other human antibody isotypes, histones, mouse and rabbit anti-biotin, mouse IgM, and biotin-tagged control for streptavidin detection. HuProt also contains landmark control spots, including Alex Fluor 555/647 (Jeong *et. al.*, 2012). Success of each batch of HuProt microarrays is validated by anti-GST staining. This confirms full-length GST-His6 fusion proteins were successfully expressed, synthesized, purified, and printed across our clone collection.

Sample Requirements

Human serum or plasma	**One 20 µL per sample**
Antibody isolates	**10 µg per sample** (concentration 0.1 mg/mL)
Human (CSF) cerebrospinal fluid	**Minimum 1.5 mL per sample**
Protein / peptide / small molecule	**10 µg per sample** (concentration 0.1 mg/mL)
RNA	**1 mL per sample** (concentration 2 µM)
Other fluids	**> Contact us**

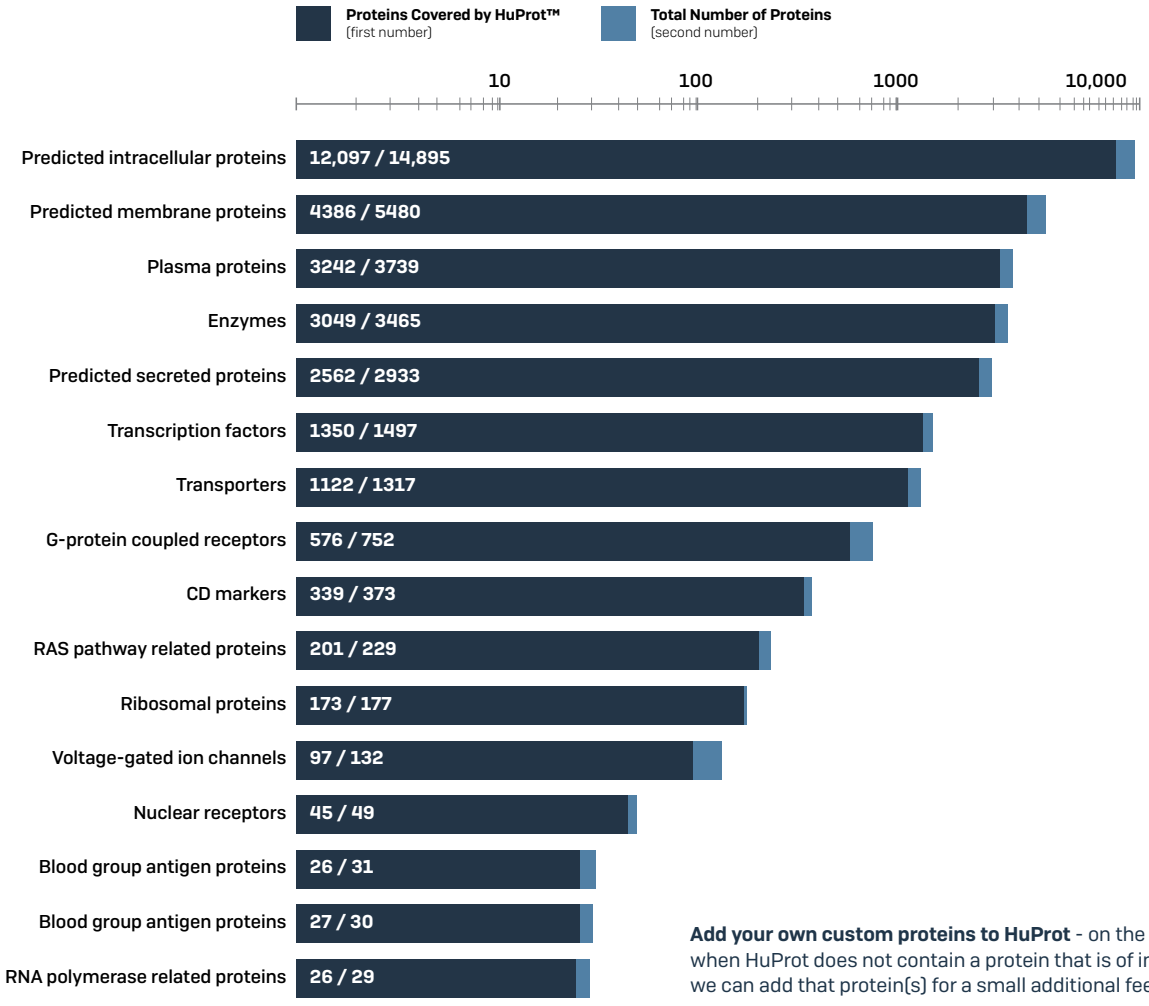
Cohort balancing (studies >60 samples)

Individual samples are run on individual HuProt slides; these are produced in batches of 600 and our lab team typically processes x60 samples per day. While data from multiple days and print batches is easily pooled by our informatics pipeline, customers running larger studies may wish to consider balancing case/control and replicate samples within and across both processing days and print batches.

HuProt™ Human Proteome Microarray

The Most Comprehensive Collection of Full-length Proteins

HuProt consists of more than 21,000 unique human proteins, isoform variants and protein fragments – covering 16,794 unique genes, it is the world's most comprehensive collection of full-length human proteins.

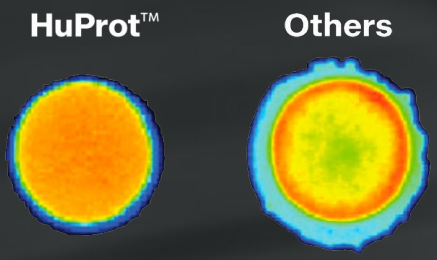


Add your own custom proteins to HuProt - on the rare occasion when HuProt does not contain a protein that is of interest to you, we can add that protein(s) for a small additional fee.

Content includes major functional classes such as intracellular proteins, membrane proteins, enzymes, secreted proteins, transcription factors, transporters, GPCRs, cytokines, immune receptors, immune checkpoints, CD markers, ion channels, cytosolic proteins, nuclear receptors. Additionally there is thorough coverage for proteins enriched in major tissues of interest such as testis, cerebral cortex, thyroid gland, skin, fallopian tube, liver, parathyroid, intestine, kidney, spleen, muscle, epididymis, lymph node, bone marrow, adrenal gland, esophagus, heart, appendix, tonsil, prostate, rectum, adipose tissue, stomach, colon, cervix, uterus, gallbladder, seminal vesicle, breast, ovary, endometrium, smooth muscle, salivary gland, pancreas, and bladder.

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Reproducible Protein Distribution



CDI Labs' non-contact piezoelectric 'inkjet' process uses next-generation Arrayjet printers and allows for rapid production of high quality microarray slides time after time. In comparison to older contact pin printing methods – HuProt arrays are made with improved accuracy and reproducibility with excellent spot morphology.

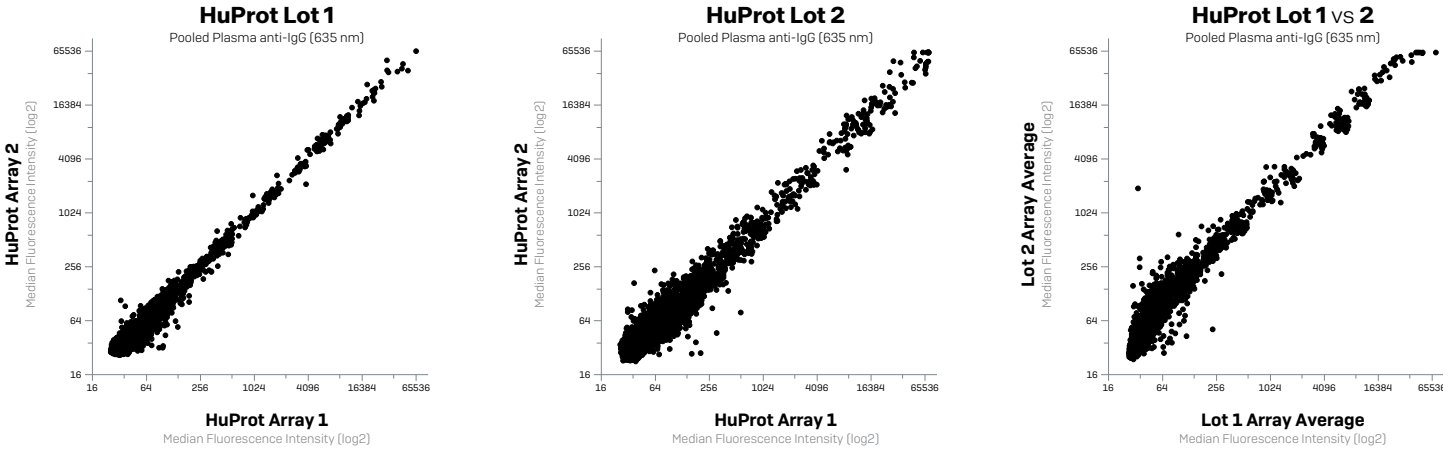
Full-length Proteins in Their Native Conformation

Protein purification prior to microarray manufacturing selects for proteins with correctly folded GST-fusion proteins. Testing on the HuProt Microarray demonstrates that the proteins are folded in native conformation and retain function. When both native and denatured HuProt microarrays were probed with monoclonal antibodies that selectively recognize either linear or folded epitopes of their cognate antigen, the antibodies were found to recognize the appropriate antigen form[2]. Further tests (RNA binding) demonstrated that proteins on HuProt do retain function.

Reproducible Serum and Plasma Profiling

Reproducible Proteome-Wide IgG Autoantibody Immunoprofiling of Healthy Human Pooled Plasma

Within and Across HuProt Proteome Microarray Batches



Pooled human plasma was incubated on pairs of HuProt proteome microarrays across two print batches and stained with a fluorescent anti-IgG secondary antibody. Raw median fluorescence intensity data were captured, log2 transformed, and plotted. Linear regression analysis demonstrates reproducibility across the array pair from Lot 1 (R^2 = 0.96), Lot 2 (R^2 = 0.92), and the averaged data from both array batches (R^2 = 0.96). These results demonstrate the robust reproducibility of HuProt autoantibody seromics data both within and across microarray production lots.



HuProt™ Focused Arrays
for Biomarker Validation

CDI Labs offers high-throughput autoantibody signature validation services using focused arrays. Autoantibody biomarkers targeting up to 300 proteins discovered using HuProt can be arrayed in a format optimized for the size of your biomarker set and throughput requirements. This service offers a fast and affordable solution for biomarker validation.

CDI Labs also synthesizes individual proteins at scale for high-throughput screening and other applications.

HuProt™ – a Transformative Product for Autoantibody Profiling

Product Feature	Benefits
~21,000 full-length human proteins	Access to the most comprehensive full-length human protein collection currently available
High-sensitivity	As little as 3 µLs of serum or plasma required
Protein purification process ensures correctly folded GST-tags	Detection of conformationally specific autoantibodies
Broad coverage across all protein subclasses	Unbiased content for true discovery applications
Single or dual color analysis available	Multi-isotype analysis on a single array
Cited 250+ publications	Performance demonstrated by the research community across many applications
Available for sale or can be processed as a service	Cost effective solutions for both small and large projects
Custom focused arrays available	Higher throughput and more cost effective for more targeted or validation experiments
Individual proteins available	A single protein (platform) can be used for discovery and highly targeted studies

VirD™ Functional Membrane Protein Array

>400 Unique Proteins for Drug Development

G protein-coupled receptors (GPCRs) play critical roles in a wide variety of physiological processes, and mutations in GPCRs are implicated in many diseases. GPCR receptors must be embedded in a cell membrane to fold correctly and are difficult to express in traditional recombinant expression systems. The most common method used to study GPCRs is to over-express them in transfected mammalian cell lines. However, the transfected cells also contain endogenous receptors, making it a challenge to interpret experimental results.

Our VirD viral-based expression technology expresses individual GPCR proteins embedded in a cell membrane. Purified virions displaying correctly-folded GPCRs are printed onto glass to create VirD arrays. This method eliminates the problem of endogenous receptor competition when using traditional methods, and yields much cleaner data. VirD GPCR arrays can be purchased directly or provided as a complete analysis service.

VirD Membrane Protein Array Validation and Production

VirD is a microarray that uses virions to display functional membrane proteins like GPCRs and is useful for drug development and functional assays.



1 Human G protein-coupled receptor (GPCR) open reading frames (ORFs) modified and cloned

2 Engineered virion clones are used to infect mammalian cells

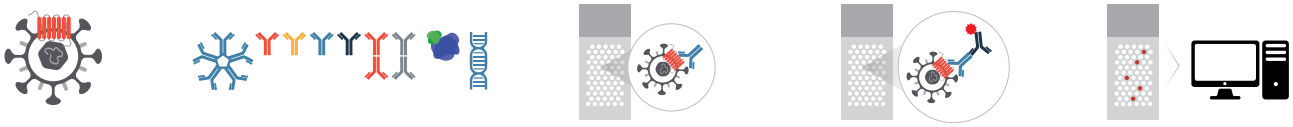
3 Transfected virions bearing V5 tags are bi-directionally sequenced

4 Arrays are printed in batches of up to 600

5 Two slides from each batch are stained for HSV-1 protein gD and scanned for QC

VirD Membrane Protein Array Methods and Analysis

VirD is a sensitive membrane protein matrix platform that provides quantitative binding interaction data from a few microliters of sample. VirD can help identify the function of orphan GPCRs and can be used to develop new drugs targeting membrane proteins and receptors.



1 Each sequence-confirmed HSV-1 virion is checked for V5 tag

2 VirD can detect most any target of interest

3 Samples of interest are diluted and reacted with array

4 Samples of interest are labeled and scanned

5 VirD library file is aligned and spot QC-ed

Technical Details

To create the VirD array, >500 unique human membrane protein open reading frames (ORFs) are modified to remove their stop codons and subcloned into the UL27 locus of the herpes simplex (HSV-1) virus genome with a C-terminal v5 tag. These individually engineered viron clones are then used to infect mammalian cells. Native human protein assembly machinery correctly folds the proteins and shuttles them to the viral membrane. Membrane-protein bearing VirD virions are concentrated from culture supernatants and their membrane-protein expression confirmed via anti-V5 immunoblots.

Purified virions were then printed on SuperEpoxy glass slides to create the final GPCR-VirD arrays. The quality of printed VirD-GPCR arrays is confirmed via staining for HSV-1 Glycoprotein D (gD) via anti-gD antibody and comparing those as compared with bovine serum albumin negative controls (BSA).

Sample Requirements

Antibody isolates	**10 µg per sample** (concentration 0.1 mg/mL)
Protein / peptide / small molecule	**10 µg per sample** (concentration 0.1 mg/mL)
RNA	**1 mL per sample** (concentration 2 µM)
Other fluids	**> Contact us**

Note

Human serum or plasma often contains anti-HSV-1 antibodies making VirD inappropriate for clinical studies.

Custom Monoclonal Antibody Development and Validation

More Than 20,000 Antigens Available

Using HuProt™ Human Proteome Microarray to validate the final product, your antibody is screened for cross-reactivity against the largest collection of human proteins on a single slide – representing approximately 75% of the human proteome.

CDI Labs' monoclonal antibody development, production and purification, along with HuProt microarray specificity validation, is (approximately) an 8-week process. It is a milestone-driven, 4-phase process with client reporting and assessment exchanged at each step. Invoicing begins at the initiation of Phase 1 and at the completion of each phase thereafter. Milestones represent potential endpoints that depend on your project requirements or if the results-to-date fail to meet your expectations.

Monospecificity in 8 Weeks

Our antibody development platform addresses the urgent need for monoclonal antibodies with unprecedented specificity to native or denatured proteins.

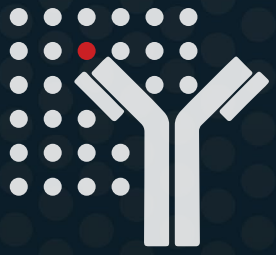
Quote	Day 1	Day 14	Day 38	Day 52	Day 56
Project Specification	**Phase 1**	**Phase 2**	**Phase 3**	**Phase 4**	**Deliverables**
Antigen design, immunization strategy and screening methodology are critical to achieving high-value results. CDI Labs offers a fully-customized development process to meet the requirements and provide a quote based on deliverables.	**Antigen and Immunization** 5-mouse immunization regimen using as little as 200 µg client-supplied protein of >80% purity or carrier-conjugated peptide. We also offer assistance in antigen design and synthesis.	**Fusion and Screening** 2 top-responding animals are chosen, and immune cells are harvested for fusion with myeloma cells yielding multiple hybridomas. Supernatants from IgG-secreting clones are screened by ELISA.	**Sub-cloning, Expansion and Specificity Testing** A minimum of 5 to a maximum of 30 high-quality clones are chosen and sub-cloned. Each is expanded into 24-well plates and subsequently into T-25 flasks. Supernatants are shipped for further client evaluation.	**Production and Purification** Each clone is vialed, labeled and frozen. Samples of each are reserved for mycoplasma testing. Per client instructions, 3 select clones are placed into antibody production. The resulting antibodies are purified and evaluated with HuProt Human Proteome Microarray.	Detailed specificity data and additional QA documentation are prepared and shipped along with the purified antibodies and all frozen clones.

Antibodies Come in a Variety of Forms Called Isotypes

Depending on the isotype, antibodies can promote cell killing, protective immune tolerance, or help educate T cells.

	IgM		IgG				IgA				IgD		IgE
Description	Prior to class switching or affinity maturation, the primary antibody response is composed of IgM. Common, with strong complement fixation.		Most abundant serum antibody. Can induce cytotoxic phagocytic responses and encourage cytotoxic T cell activation via targeting to antigen presenting cells. Can be tolerogenic depending on subclass when bound to FcγRIIB. Crosses placenta.				Dimeric form of IgA2 is the most abundant secreted antibody across mucous membranes such as in the mouth and colon as well as breast milk. Monomers of IgA1 are more common in serum.				Membrane-bound IgD is a hallmark of mature B cells. Found in serum in low amounts, no consensus on exact serum function		Rare antibody with extreme ability to activate FcεR allergic responses. Evolved for parasite defense.
	 IgM B cell Receptor	 IgM Pentamer	 IgG1	 IgG2	 IgG3	 IgG4	 Secreted IgA1 Dimer	 IgA1	 Secreted IgA2 Dimer	 IgA2	 IgD B cell Receptor	 IgD	 IgE
Approximate % of total serum antibody	0	10%	53%	15%	6%	4%	0	11%	0	1%	0	0.2%	0.004%
Approximate serum half-life (days)	0	5–10	21	21	7	21	0	5–7	0	4–6	0	2–8	1–5
Often cytotoxic activation by immune complex FcγRIIA	–	–					–	–	–	–	–	–	–
Often tolerogenic inhibition by immune complex FcγRIIB	–	–		–			–	–	–	–	–	–	–
Activation of classical complement pathway (C1q)	–					–	–		–	–	–	–	–
Ability to cross the placenta during pregnancy	–	–					–	–	–	–	–	–	–
Dominant presence in breast milk and mucous membranes	–	–	–	–	–	–		–		–	–	–	–
Single molecule FcεR activation; response to parasites and allergens	–	–	–	–	–	–	–	–	–	–	–	–	

Chart prepared by Tyler Hulett, PhD *with references (1-3). References* ***1.*** *Nimmerjahn, F. Divergent Immunoglobulin G Subclass Activity Through Selective Fc Receptor Binding. Science 310, 1510-1512 (2005).*
2. *Vidarsson, G. Dekkers, G & Rispens, T. IgG Subclasses and Allotypes From Structure to Effector Functions. Front. Immuno. 5, (2014).* ***3.*** *Fundamental Immunology. (Walters Kluwer/Uppincott Williams & Wilkins, 2008).*



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